# ARJUNA JEE 2027

Lecture -05 | Mathematics

# SEQUENCE & SERIES

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# Topics *to be covered*

A **Different types of G.P.**

B **Harmonic Progression (H.P.)**

C **Question Practice**

D



# Question Practice

## Question [JEE Mains-2023]

Re-attempt

61%

For three positive integers $p, q, r$, $x^{pq^2} = y^{qr} = z^{p^2r}$ and $r = pq + 1$ such that $3, 3\log_y x, 3\log_z y, 7\log_x z$ are in A.P. with common difference 1/2. Then $r - p - q$ is equal to

7 - 2 - 3

(A) 2

(B) 6

(C) 12

(D) −6

# $x^{pq^2} = y^{qr} = z^{p^2r} = (t)$

$$x = (t)^{\frac{1}{pq^2}},\ y = (t)^{\frac{1}{qr}},\ z = (t)^{\frac{1}{p^2r}}$$

$$\log_y x = \log_{(t)^{\frac{1}{qr}}} (t)^{\frac{1}{pq^2}} = \frac{qr}{pq^2}$$

$$\log_z y = \frac{p^2r}{qr},\quad \log_x z = \frac{pq^2}{p^2r}$$

$3,\ 3\frac{qr}{pq^2},\ 3\frac{p^2r}{qr},\ 7\frac{pq^2}{p^2r}$ → A.P.

$3,\ 3\frac{r}{pq},\ 3\frac{p^2}{q},\ \frac{7q^2}{pr}$ → A.P.

$3, \frac{3r}{pq}, \frac{3p^2}{q}, \frac{7q^2}{pr}$ → A.P.

$+\frac{1}{2}$ $+\frac{1}{2}$

$\frac{1}{2} = \frac{3r}{pq} - 3 = \frac{3p^2}{q} - \frac{3r}{pq}$

$3 + \frac{1}{2} = \frac{3r}{pq}$

$\frac{7}{2} = \frac{3r}{r-1}$

$7r - 7 = 6r$

$r = 7$

$r = pq + 1$

$r - 1 = pq$

$6 = pq$

$q = \frac{6}{p}$

$q = 3$

# $\frac{3p^2}{q} - 3 = 1$

$\frac{3p^2}{q} = 4$

$3p^2 = 4q$

$3p^2 = 4 \times \frac{6}{p}$ → $p^3 = 8$

$p = 2$

## Question [JEE Mains-2024]

**An arithmetic progression is written in the following way**

$R_1 \longrightarrow 2$

$R_2 \longrightarrow 5 \quad 8$

$R_3 \longrightarrow 11 \quad 14 \quad 17$

$R_4 \longrightarrow 20 \quad 23 \quad 26 \quad 29$

Pattern - observ.

A.P. $\leftarrow \{2, 5, 8, 11, 14, 17, 20, 23, 26, 29 \ldots\}$

**The sum of all the terms of the 10th row is _____.**

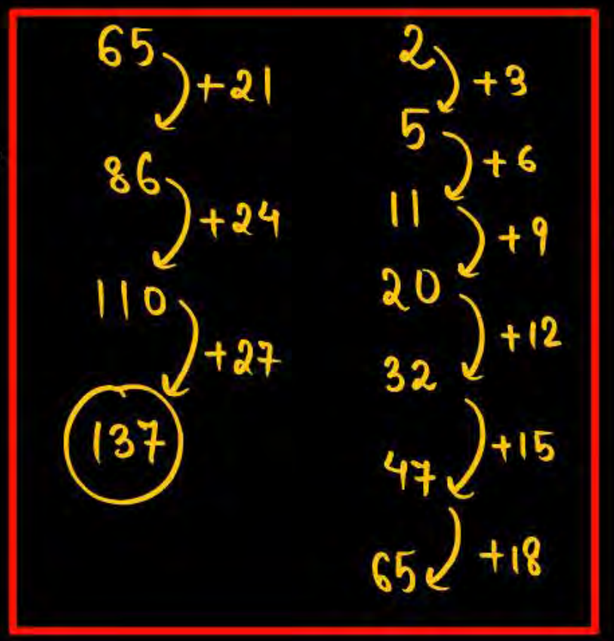
10 elem.

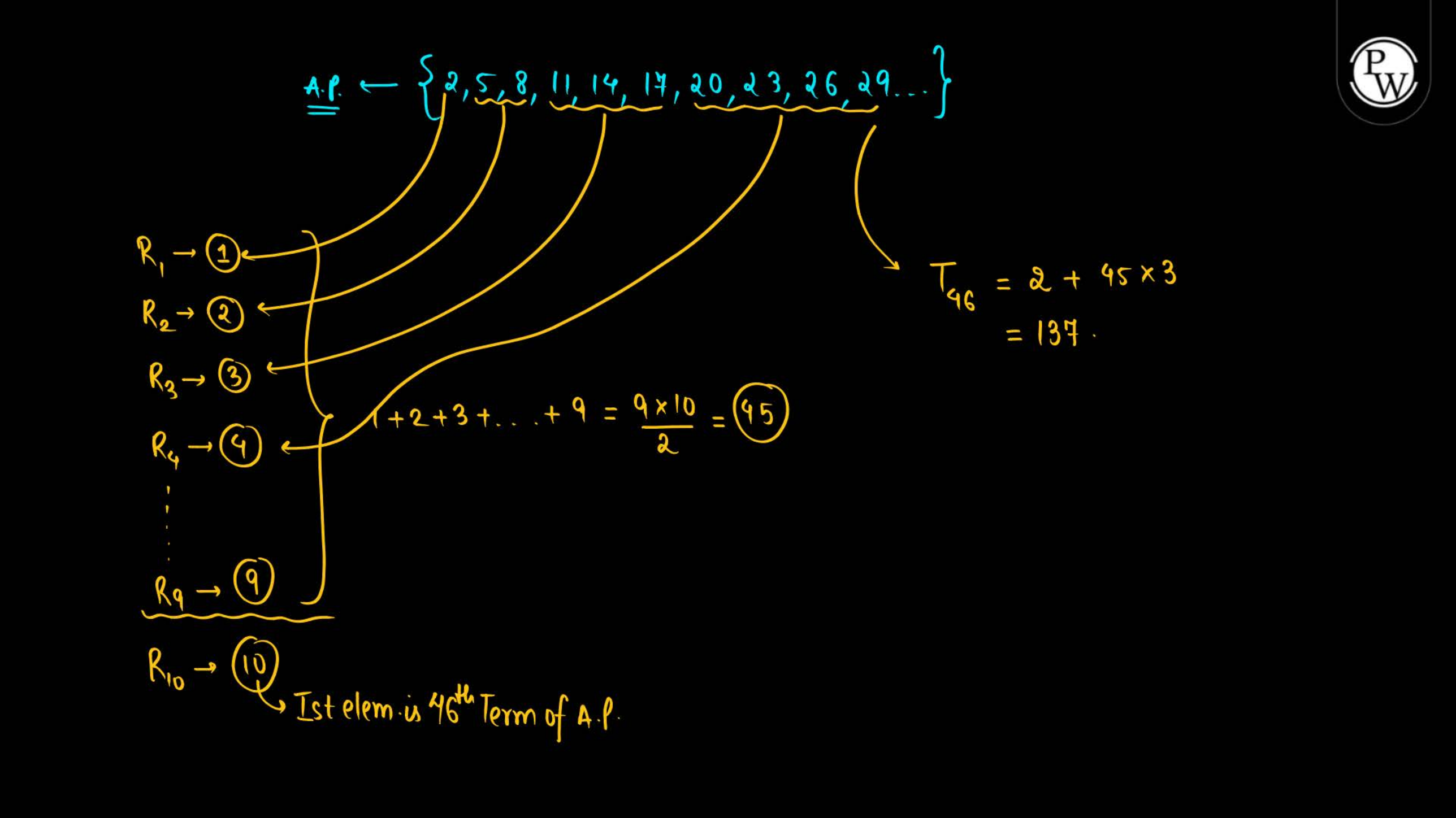
$R_{10} \rightarrow 137, 140, \ldots\ldots\ldots$

$d = 3.$

$\text{Sum} = \frac{10}{2}\{2(137) + (10-1)3\}$

$= 1505.$




A.P. ← {2, 5, 8, 11, 14, 17, 20, 23, 26, 29...}
R1 → 1
R2 → 2
R3 → 3
R4 → 4
R9 → 9
R10 → 10
1+2+3+...+9 = 9×10/2 = 45
T46 = 2 + 45×3
= 137.
1st elem. is 46th Term of A.P.

## Question [JEE Mains-2023]

**Ans: (8)**

Let $a_1, a_2, \ldots\ldots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$, then $12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}}+\frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}}+\ldots+\frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$ is equal to

$$\frac{1}{\sqrt{a_{11}}+\sqrt{a_{10}}} \times \frac{\sqrt{a_{11}}-\sqrt{a_{10}}}{\sqrt{a_{11}}-\sqrt{a_{10}}} = \frac{\sqrt{a_{11}}-\sqrt{a_{10}}}{a_{11}-a_{10}}$$

$d$

$$\frac{1}{\sqrt{a_{12}}+\sqrt{a_{11}}} \times \frac{\sqrt{a_{12}}-\sqrt{a_{11}}}{\sqrt{a_{12}}-\sqrt{a_{11}}} = \frac{\sqrt{a_{12}}-\sqrt{a_{11}}}{a_{12}-a_{11}}$$

$d$

$$\frac{\sqrt{a_{13}}-\sqrt{a_{12}}}{a_{13}-a_{12}}$$

$d$

## Question [JEE Mains-2023]

**Ans: (8)**

Let $a_1, a_2, \ldots\ldots, a_n$ be in A.P. If $a_5 = 2a_7$ and $a_{11} = 18$ then $12\left(\frac{1}{\sqrt{a_{10}}+\sqrt{a_{11}}}+\frac{1}{\sqrt{a_{11}}+\sqrt{a_{12}}}+\ldots+\frac{1}{\sqrt{a_{17}}+\sqrt{a_{18}}}\right)$ is equal to

$$12\times\frac{1}{d}\left\{\left(\sqrt{a_{11}}-\sqrt{a_{10}}\right)+\left(\sqrt{a_{12}}-\sqrt{a_{11}}\right)+\left(\sqrt{a_{13}}-\sqrt{a_{12}}\right)+\ldots+\left(\sqrt{a_{18}}-\sqrt{a_{17}}\right)\right\}$$

$$\#\ \text{Inf.} = 12\left(\frac{\sqrt{a_{18}}-\sqrt{a_{10}}}{d}\right) = 8$$

OR

$$\begin{aligned}&\sqrt{a_{11}}-\sqrt{a_{10}}\\ +\ &\sqrt{a_{12}}-\sqrt{a_{11}}\\ +\ &\sqrt{a_{13}}-\sqrt{a_{12}}\\ &\vdots\\ +\ &\sqrt{a_{18}}-\sqrt{a_{17}}\\ \hline &\sqrt{a_{18}}-\sqrt{a_{10}}\ \checkmark\end{aligned}$$



# GEOMETRIC PROGRESSION

$$S = a + ar + ar^2 + ar^3 + \ldots\ldots\ldots\ldots \text{ upto } n - \text{terms}$$

$$S_n = a\left(\frac{r^n - 1}{r - 1}\right), \quad (r \neq 1)$$

$$S_n = na, \quad (r = 1)$$

$$S_\infty = \frac{a}{1 - r}, \quad -1 < r < 1$$

## Question

The square of product of $2^{\frac{1}{4}} \cdot 4^{\frac{1}{16}} \cdot 8^{\frac{1}{48}} \cdot 16^{\frac{1}{128}} \cdot \ldots\ldots$ to $\infty$ is equal to

$$= 2^{\left(\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\ldots\infty\right)} = 2^{\frac{1}{2}} = \sqrt{2}$$

$2^{\frac{1}{4}} \cdot 2^{\frac{1}{8}} \cdot 2^{\frac{1}{16}} \cdot 2^{\frac{1}{32}} \ldots\ldots \infty$

$\infty$ - G.P.

$a = \frac{1}{4}$, $r = \frac{1}{2}$

$$S_\infty = \frac{\frac{1}{4}}{1-\frac{1}{2}} = \frac{\left(\frac{1}{4}\right)}{\left(\frac{1}{2}\right)} = \frac{2}{4} = \frac{1}{2}$$

$4^{\frac{1}{16}} = \left(2^2\right)^{\frac{1}{16}} = 2^{\frac{1}{8}}$

$16^{\frac{1}{128}} = \left(2^4\right)^{\frac{1}{32}} = 2^{\frac{1}{32}}$

$8^{\frac{1}{48}} = \left(2^3\right)^{\frac{1}{16}} = 2^{\frac{1}{16}}$

## Question [JEE Mains-2023]

The 4th term of GP is 500 and its common ratio is $\frac{1}{m}$, $m \in \mathbb{N}$. Let $S_n$ denote the sum of the first $n$ terms of this GP. If $S_6 > S_5 + 1$ and $S_7 < S_6 + \frac{1}{2}$, then the number of possible values of $m$ is equal to

# $T_4 = 500$.

$r = \frac{1}{m}$, $m \in N$.

$a\left(\frac{1}{m}\right)^3 = 500$.

$\frac{a}{m^3} = 500$

$a = 500m^3$.

# $S_6 - S_5 > 1$

# $T_6 > 1$

$a\left(\frac{1}{m}\right)^5 > 1$

$\frac{a}{m^5} > 1$

$\frac{500m^3}{m^5} > 1$ → $500 > m^2$

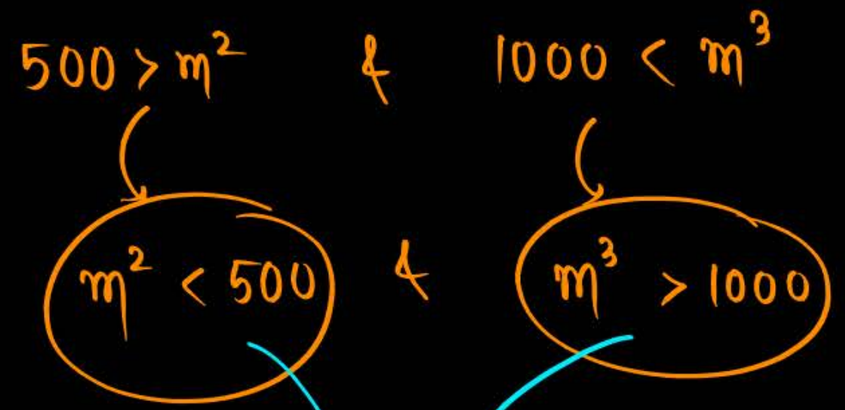
$S_7 - S_6 < \frac{1}{2}$

$T_7 < \frac{1}{2}$

$a\left(\frac{1}{m}\right)^6 < \frac{1}{2}$

$\frac{500m^3}{m^6} < \frac{1}{2}$

$1000 < m^3$




$n = \{11, 12, 13, 14, 15 \ldots\ldots 20, 21, 22\}$

Natural no. whose cube is greater than 1000 but sq. is less than 500.

no. of values $= 22 - 11 + 1$

$= 11 + 1$

$= 12$ Ans.

## Question [JEE Mains-2024]

Let $a, ar, ar^2, \ldots\ldots,$ be an infinite G.P. If $\sum_{n=0}^{\infty} ar^n = 57$ and $\sum_{n=0}^{\infty} a^3 r^{3n} = 9747$, then $a + 18r$ is equal to

$-1 < r < 1$

$\sum_{n=0}^{\infty} ar^n = 57$

$a + ar^1 + ar^2 \ldots\ldots \infty = 57$

$\frac{a}{1-r} = 57$ — (1)

$\sum_{n=0}^{\infty} a^3 r^{3n} = a^3 r^0 + a^3 r^3 + a^3 r^6 + \ldots\ldots \infty$

$= a^3 + a^3 r^3 + a^3 r^6 + \ldots\ldots \infty$

$\frac{a^3}{1-r^3} = 9747$ — (2)

$\frac{a^3}{1-r^3} = 9747$

$19 + \frac{6}{\cancel{18}} \times \frac{2}{\cancel{3}} = 31$

A. 27

B. 46

C. 38

D. 31

$\frac{a}{1-r} = 57$ — (1)

$a = 57(1-r)$

$\frac{a^3}{1-r^3} = 9747$ — (2)

$a = 57\left(\frac{1}{3}\right)$

$a = 19$

$\frac{57^3(1-r)^3}{1^3-r^3} = 9747 = 171 \times 57$

$\frac{(1-r)^3}{(1-r)(1+r+r^2)} = \frac{57}{171} \times \frac{57}{57 \times 57 \times 57}$

$19$

$19(1+r^2-2r) = r^2+r+1$

$18r^2 - 39r + 18 = 0$

$18r^2 - 27r - 12r + 18 = 0$

$9r(2r-3) - 6(2r-3) = 0$

$(9r-6)(2r-3) = 0$

$r = \frac{3}{2}, \frac{6}{9} = \frac{2}{3}$

Rejected.

# FORMAT

# Ek ki power badh rhi hai, dusre ki power kum ho rhi hai.

$$a^n + b\,a^{n-1} + b^2\,a^{n-2} + \ldots\ldots + b^{n-1}\,a + b^n$$

**First Term =** $a^n$

**Common Ratio =** $\frac{b}{a}$

**Number of Terms =** $(n+1)$

# G.P. #

$$\text{Sum} = a^n \left( \frac{\left(\frac{b}{a}\right)^{n+1} - 1}{\left(\frac{b}{a}\right) - 1} \right) = \frac{b^{n+1} - a^{n+1}}{b - a}$$

RAR

# FORMAT

**# Ek ki power badh rhi hai, dusre ki power kum ho rhi hai.**

$$a^n - b\,a^{n-1} + b^2\,a^{n-2} - \ldots\ldots \quad (n+1)\text{ terms}$$

**First Term =** $a^n$

**Common Ratio =** $\frac{-b}{a}$

**Number of Terms =** $(n+1)$

$$\text{Sum} = \frac{(-b)^{n+1} - a^{n+1}}{(-b) - a}$$

$$\text{Sum} = a^n \left( \frac{\left(\frac{b}{a}\right)^{n+1} - 1}{\left(\frac{b}{a}\right) - 1} \right)$$

$$= \frac{a^{n+1}}{a^n} \left( \frac{b^{n+1} - a^{n+1}}{\frac{b - a}{a}} \right) = \frac{b^{n+1} - a^{n+1}}{b - a}$$

# G.P. ⟶ ex:- $T_1$ $T_2$ $T_3$ $T_4$

$2^3$, $2^2 \cdot 3^1$, $2^1 \cdot 3^2$, $3^3$

$a = 2^3$, $r = \frac{3}{2}$, $n = 4$.

$$\frac{T_2}{T_1} = \frac{2^2 \cdot 3^1}{2^2} = \frac{3}{2}$$

$$\frac{T_3}{T_2} = \frac{2^1 \cdot 3^2}{2^2 \cdot 3^1} = \frac{3}{2}$$

$$\frac{T_4}{T_3} = \frac{3^3}{2^1 \cdot 3^2} = \frac{3}{2}$$

## RAPID FIRE ROUND

(i) $S = 3^4 + 3^3 \cdot 2^1 + 3^2 \cdot 2^2 + 3^1 \cdot 2^3 + 2^4$

(ii) $S = 5^6 - 5^5 \cdot 3^1 + 5^4 \cdot 3^2 - 5^3 \cdot 3^3 + 5^2 \cdot 3^4 - 5^1 \cdot 3^5 + 3^6$

$a = 3^4,\ r = \frac{2}{3},\ n.o.t. = 5$

$\# S = \frac{2^5 - 3^5}{2 - 3} = 3^5 - 2^5$

$a = 5^6,\ r = -\frac{3}{5},\ n.o.t = 7$

$S = 5^6 \left( \frac{\left(-\frac{3}{5}\right)^7 - 1}{\left(-\frac{3}{5}\right) - 1} \right) = \frac{5^6}{5^7} \left( \frac{-3^7 - 5^7}{-3 - 5} \right)$

$= \left( \frac{3^7 + 5^7}{8} \right)$ Ans.

## Question [JEE Mains-2020]

If $2^{10} + 2^9 \cdot 3^1 + 2^8 \cdot 3^2 + \ldots\ldots + 2 \cdot 3^9 + 3^{10} = S - 2^{11}$, then $S$ is equal to

A $2 \cdot 3^{11}$

B $3^{11} - 2^{12}$

C $\frac{3^{11}}{2} + 2^{10}$

D $3^{11}$

# Q iBY - (X)

## Question [JEE Mains-2020]

If $|x| < 1, |y| < 1$, and $x \neq y$, then the sum to infinity of the following series $(x+y) + (x^2+xy+y^2) + (x^3+x^2y+xy^2+y^3) + \ldots\ldots$ is

A $\dfrac{x+y+xy}{(1+x)(1+y)}$

B $\dfrac{x+y-xy}{(1-x)(1-y)}$

C $\dfrac{x+y-xy}{(1+x)(1+y)}$

D $\dfrac{x+y+xy}{(1-x)(1-y)}$

$T_1 = x+y = \dfrac{x^2-y^2}{x-y}$

$T_2 = x^2+xy+y^2 = \dfrac{x^3-y^3}{x-y}$

$T_3 = x^3+x^2y+xy^2+y^3 = \dfrac{x^4-y^4}{x-y}$

$\vdots$

$\infty$

add

$S_\infty = \dfrac{1}{(x-y)}\left\{x^2-y^2+x^3-y^3+x^4-y^4+\ldots\infty\right\}$

$\left(\dfrac{1}{x-y}\right)\left\{(x^2+x^3+x^4+\ldots\infty)-(y^2+y^3+y^4+\ldots\infty)\right\}$

$S_\infty = \left(\dfrac{1}{x-y}\right)\left\{\left(\dfrac{x^2}{1-x}\right)-\left(\dfrac{y^2}{1-y}\right)\right\}$

$$S_\infty = \left(\frac{1}{x-y}\right)\left\{\left(\frac{x^2}{1-x}\right) - \left(\frac{y^2}{1-y}\right)\right\}$$

$$S_\infty = \frac{1}{(x-y)} \cdot \frac{(x^2 - x^2y - y^2 + xy^2)}{(1-x)(1-y)}$$

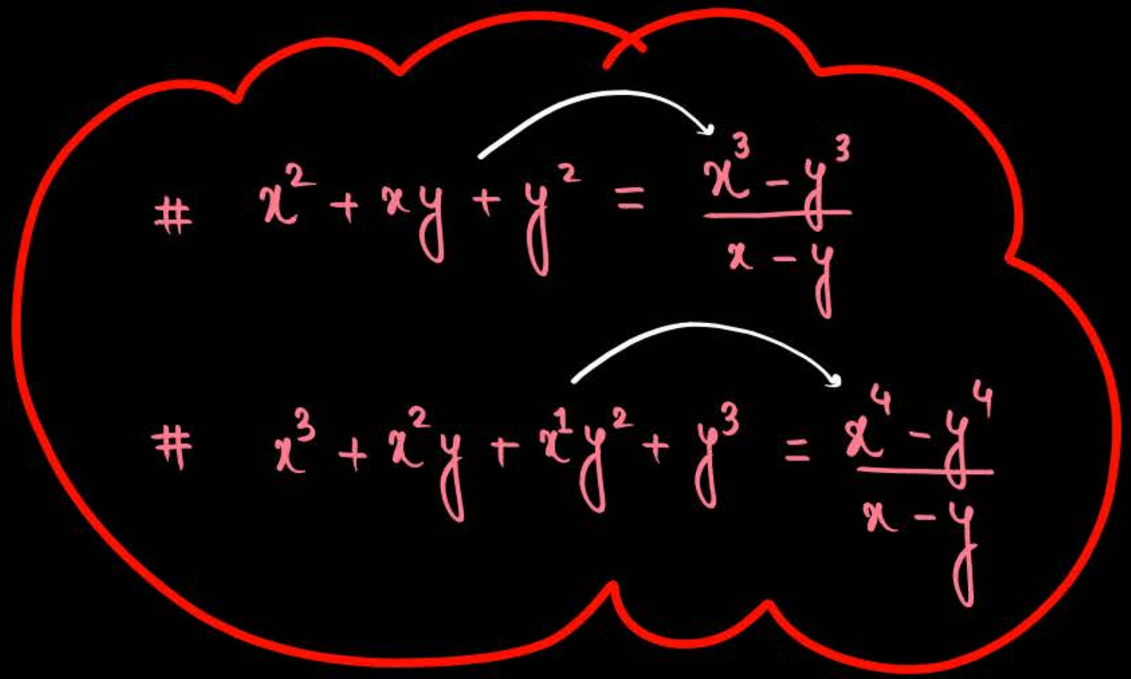
$$x^2 - y^2 + xy^2 - x^2y$$

$$(x-y)(x+y) - xy(x-y)$$

$$(x-y)(x+y-xy)$$

$$S_\infty = \frac{1}{(x-y)} \cdot \frac{(x-y)(x+y-xy)}{(1-x)(1-y)}$$

$$S_\infty = \frac{x+y-xy}{1-x-y+xy}$$


# $x^2 + xy + y^2 = \frac{x^3 - y^3}{x - y}$
# $x^3 + x^2y + xy^2 + y^3 = \frac{x^4 - y^4}{x - y}$

# RAPID FIRE ROUND ON HIDDEN G.P.

(i) $S = 9 + 99 + 999 + 9999$

(ii) $S = 5 + 55 + 555 + 5555$

$S = 5(1 + 11 + 111 + 1111)$

$S = \frac{5}{9}(9 + 99 + 999 + 9999)$

(iii) $x = 0.4\overline{18}$, then $x$ = _____.

(iv) Show that $n$-digit number whose each digit is $\alpha = \frac{\alpha}{9}(10^n - 1)$.

$S = 10 - 1 + 100 - 1 + 1000 - 1 + 10000 - 1$

$S = \underbrace{(10^1 + 10^2 + 10^3 + 10^4)}_{\text{G.P.}} - 4$

(3) $x = 0.4\overline{18}$

$x = 0.418181818\ldots\infty$

$10 \times$ → $10x = 4.181818\ldots$

$1000 \times$ → $1000x = 418.181818\ldots$

$10x = 4.181818\ldots$

$990x = 414.0$

$x = \frac{414}{990}$

(3)

$$x = 0.4181818\ldots\infty$$

$$x = 0.4 + 0.018 + 0.00018 + 0.0000018 + \ldots\infty$$

$$= 0.4 + \left(\frac{18}{10^3} + \frac{18}{10^5} + \frac{18}{10^7} + \ldots\infty\right)$$

$$= 0.4 + 18\left(\frac{1}{10^3} + \frac{1}{10^5} + \frac{1}{10^7} + \frac{1}{10^9} + \ldots\infty\right)$$

G.P.

# RAPID FIRE ROUND ON HIDDEN G.P.

(i) $S = 9 + 99 + 999 + 9999$

(ii) $S = 5 + 55 + 555 + 5555$

(iii) $x = 0.4\overline{18}$, then $x$ = _____.

(iv) Show that $n$-digit number whose each digit is $\alpha = \frac{\alpha}{9}(10^n - 1)$.

$\underbrace{\alpha\alpha\alpha\alpha\ldots\alpha}_{n\text{-Times}} = \alpha \times 10^{n-1} + \ldots + \alpha \times 10^2 + \alpha \times 10^1 + \alpha \times 1$

$= \alpha\left(10^{n-1} + 10^{n-2} + \ldots + 10^2 + 10^1 + 1\right)$

$= \alpha\left(\frac{10^n - 1}{9}\right)$

$\underline{4}\underline{2}\underline{3}\underline{1} = 4\times10^{3} + 2\times10^{2} + 3\times10 + 1\times1$

# expanded

$77777 = 7\times10^{4} + 7\times10^{3} + 7\times10^{2} + 7\times10 + 7\times1$

$= 7\left(10^{4} + 10^{3} + 10^{2} + 10^{1} + 1\right)$

$7 \times 10^{4}\left(\frac{1-\left(\frac{1}{10}\right)^{5}}{1-\frac{1}{10}}\right) = 7\times\frac{10^{4}}{10^{5}}\times\frac{10^{5}-1}{\left(\frac{9}{10}\right)} = 7\times\left(\frac{10^{5}-1}{9}\right)$

## Question [JEE Mains-2013]

**The sum of first 20 terms of the sequence 0.7, 0.77, 0.777, .... is**

A $\frac{7}{81}(179 - 10^{-20})$

B $\frac{7}{9}(99 - 10^{-20})$

C $\frac{7}{81}(179 + 10^{-20})$

D $\frac{7}{9}(99 + 10^{-20})$

$$S = \frac{7}{10} + \frac{77}{100} + \frac{777}{1000} + \text{-----20 Terms.}$$

$$= \frac{7}{9}\left(\frac{10-1}{10} + \frac{100-1}{100} + \frac{1000-1}{1000} + \text{------}\right)$$

$$= \frac{7}{9}\left(\frac{10}{10}\frac{-1}{10} + \frac{100}{100}\frac{-1}{100} + \frac{1000}{1000}\frac{-1}{1000} + \text{----}\right)$$

$$= \frac{7}{9}\left(1 - \frac{1}{10} + 1 - \frac{1}{100} + 1 - \frac{1}{1000} + \text{---}\right)$$

$$S = \frac{7}{9}\left(1 - \frac{1}{10} + 1 - \frac{1}{100} + 1 - \frac{1}{1000} + \ldots\ldots\ldots\right)$$

$$S = \frac{7}{9}\left(20 - \left(\frac{1}{10} + \frac{1}{100} + \frac{1}{1000} + \ldots\ldots \text{ 20 Terms}\right)\right)$$

$20 - \frac{1}{9}$

$$\left(\frac{1}{10}\right)\left\{\frac{1 - \left(\frac{1}{10}\right)^{20}}{1 - \left(\frac{1}{10}\right)}\right\} = \frac{1}{10}\left(\frac{10^{20} - 1}{\frac{9}{10} \times 10^{20}}\right)$$

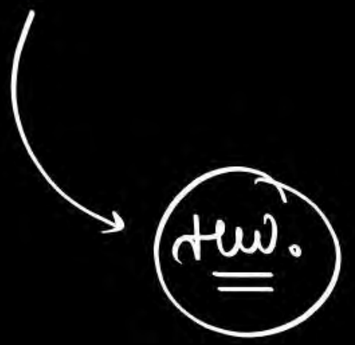
$$= \frac{1}{9}\left(\frac{10^{20} - 1}{10^{20}}\right) = \frac{1}{9}\left(1 - 10^{-20}\right)$$

$$S = \frac{7}{9}\left(20 - \frac{1}{9}\left(1 - 10^{-20}\right)\right)$$

$$= \frac{7}{9}\left(\frac{180 - 1 + 10^{-20}}{9}\right) = \frac{7}{81}\left(179 + 10^{-20}\right) \text{ Ans}$$

## Acidic Question

Find the sum of series $\frac{3}{19} + \frac{33}{19^2} + \frac{333}{19^3} + \frac{3333}{19^4} + \ldots\ldots \infty$



# DO IT BY YOURSELF (DIBY)

# DIBY-24

Let $a_1, a_2, \ldots a_{2024}$ be an Arithmetic Progression such that $a_1 + (a_5 + a_{10} \quad + a_{15} + \cdots + a_{2020}) + a_{2024} = 2223$. Then $a_1 + a_2 + a_3 + \cdots + a_{2024}$ is equal to

[JEE Mains-2025]

[Ans. : 11132]

# DIBY-25

If $p^{th}$, $q^{th}$ and $r^{th}$ term of an A.P. are respectively a, b and c, then the value of $a(q-r) + b(r-p) + c(p-q)$ is

(A) 0 (B) 1 (C) $p+q+r$ (D) $abc$

[Ans. : (A)]

# DIBY-26

Find the value of the sum $S = 100^2 - 99^2 + 98^2 - 97^2 + 96^2 - 95^2 + \ldots + 2^2 - 1^2$

(A) 4950 (B) 10100 (C) 5150 (D) 5050

[Ans. : (D)]

# DIBY-27

The sum of first 3 terms of an A.P. is 27 and the sum of their squares is 293, then first term and common difference of A.P. is

(A) First term $= 3$, Common difference $= \pm 6$

(B) First term = 4, Common difference = 5

(C) First term = 4 or 14, Common difference = $\pm 5$

(D) First term = 2 or 16, Common difference $= \pm 7$

[Ans. : (C)]

# DIBY-28

The number of terms in an A.P. is even, the sum of the odd numbered terms is 24 & sum of the even numbered terms is 30 and the last term exceeds the first term by $10\frac{1}{2}$. Find the number of terms.

(A) 4 (B) 6 (C) 8 (D) 12

[Ans. : (C)]

# DIBY-29

How many terms of the series, $20 + 19\frac{1}{3} + 18\frac{2}{3} + \cdots ..$ must be taken so that sum is 300. Explain the reason of double answer.

(A) $n = 25$ or $n = 36$; the sum of the terms from $26^{th}$ to $36^{th}$ is zero.

(B) $n = 20$ or $n = 30$; the common difference becomes positive after the $20^{th}$ term.

(C) $n = 15$ or $n = 40$; the terms after the $15^{th}$ term are perfectly symmetrical.

(D) $n = 25$ only; the negative value of $n = -36$ is rejected.

[Ans. : (A)]

# DIBY-30

If the sum of the roots of the equation $ax^2 + bx + c = 0$ is equal to the sum of the squares of their reciprocals, then show that $bc^2, ca^2, ab^2$ are in A.P.

# DIBY-31

A person is to count 4500 currency notes. Let $a_n$ denote the number of notes he counts in the $n^{th}$ minute. If $a_1 = a_2 = \cdots = a_{10} = 150$ and $a_{10}, a_{11}, \ldots$ are in an AP with common difference -2, then the time taken by him to count all notes is

[AIEEE-2010]

(A) 24 minutes (B) 34 minutes (C) 125 minutes (D) 135 minutes

[Ans. : (B)]

# DIBY-32

Let the sum of the first $n$ terms of a non-constant A.P., $a_1, a_2, a_3 \ldots$ be $50n + \frac{n(n-7)}{2}A$, where $A$ is constant. If $d$ is the common difference of this A.P., then the ordered pair $(d, a_{50})$ is equal to:

[JEE Mains-2019]

(A) $(A, 50 + 45A)$ (B) $(50, 50 + 46A)$

(C) $(A, 50 + 46A)$ (D) $(50, 50 + 45A)$

[Ans. : (C)]

# DIBY-33

Along a road lies an odd number of stones placed at intervals of 10 m. These stones have to be assembled around the middle stone. A person can carry only one stone at a time. A man carried out the job starting with the stone in the middle, carrying stones in succession, thereby covering a distance of 4.8 km. Then the number of stones is

(A) 15 (B) 29 (C) 31 (D) 35

[Ans. : (C)]

# GEOMETRIC PROGRESSION

**# DIBY-34**

Let $a_1, a_2, a_3, \ldots$ be a G. P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{11T}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to [JEE Mains-2025]

(A) 131 (B) 130 (C) 129 (D) 128

[Ans. : (C)]

**# DIBY-35**

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms. If $a_1 a_5 = 28$ and $a_2 + a_4 = 29$, then $a_6$ is equal to: [JEE Mains-2025]

(A) 812 (B) 526 (C) 784 (D) 628

[Ans. : (C)]

**# DIBY-36**

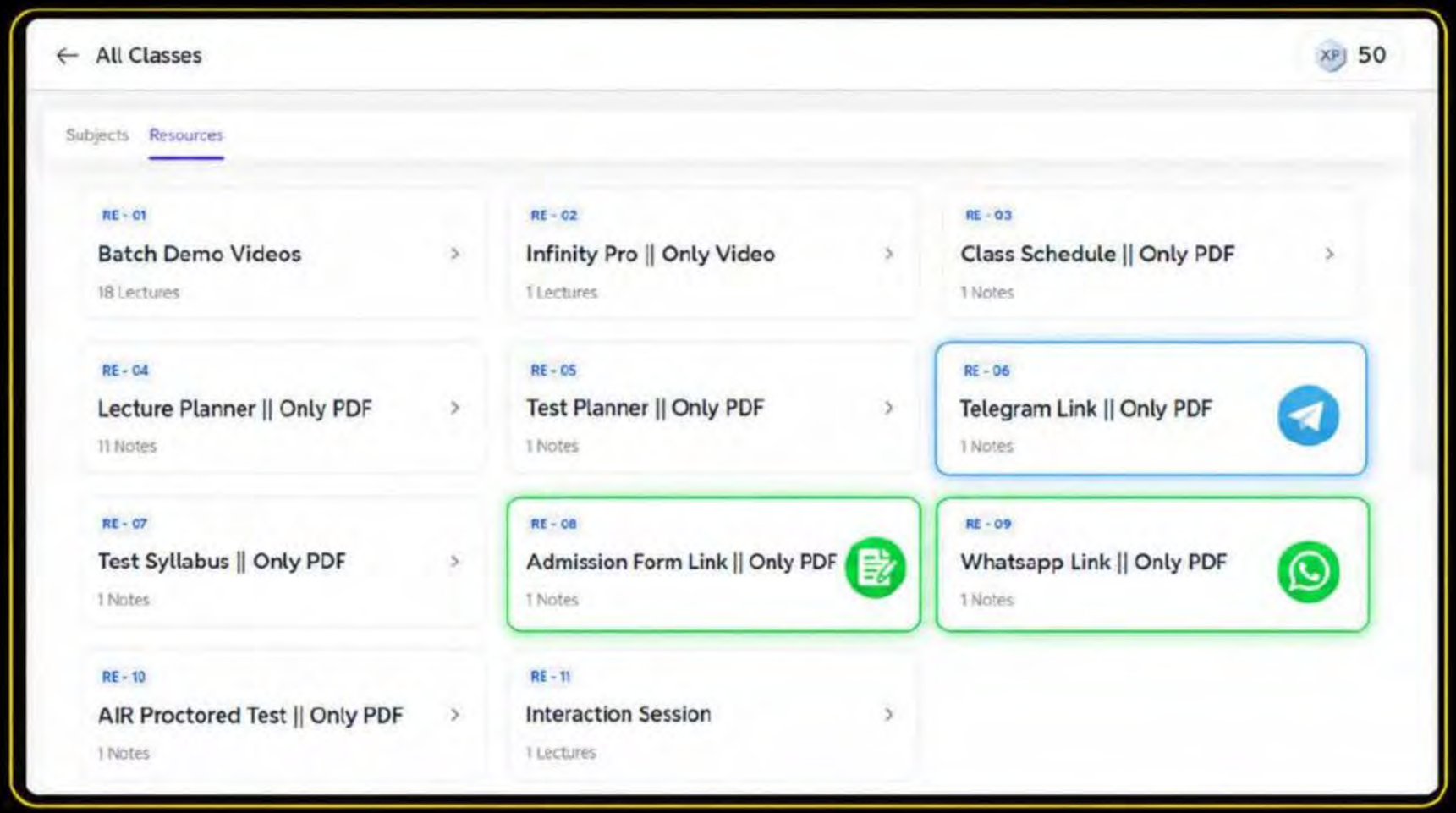
Let $a$ and $b$ be two distinct positive real numbers. Let 11$^{th}$ term of a $GP$, whose first term is $a$ and third term is $b$, is equal to $p^{th}$ term of another $GP$, whose first term is $a$ and fifth term is $b$. Then $p$ is equal to [JEE Mains-2024]

(A) 20 (B) 25 (C) 21 (D) 24

[Ans. : (C)]

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*It is never about end result*

*It's always about Journey*

# Thank You

#futureIITians